Hong Kong Metropolitan University
COMP2090SEF
Data Structures, Algorithms And Problem Solving
Course Project Report
Group 11
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We declare that:
(i) all members of the group have read and checked that all parts of the project (including proposal, code programs, and reports), irrespective of whether they are contributed by individual member or all members as a group, here submitted is original except for source material explicitly acknowledged;
(ii) the project, in parts or in whole, has not been submitted for more than one purpose without declaration;
(iii) we are aware of the University's policy and regulations on honesty in academic work and understand the possible consequence when breaching such policy and regulations;
(iv) we confirm that we have declared in the report about the usage of AI and other generative models, including but not limited to ChatGPT, LLaMA, Gemini, Mistral, and Stable Diffusion, and complied with the instructions provided by HKMU; and
(v) we are aware that all members of the group should be held responsible and liable to disciplinary actions, irrespective of whether he/she has signed the declaration and whether he/she has contributed, directly or indirectly, to the problematic contents.
We confirm that we have read through and understood the project requirements. We understand that failure to comply with the project requirements will result in score deduction.

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1. Project Overview

This project is a Student Score Management System developed based on Object-Oriented Programming (OOP) concepts, which aims to solve the real-life problem of student information and score management for teachers. The system provides a complete set of functions to manage student basic information, course scores, score ranking and statistical analysis, helping teachers to efficiently process student academic data. The project is implemented with 3 independent Python modules, which meets the requirement of modular programming, and fully covers all OOP concepts introduced in the course to ensure the code is maintainable, extensible and robust.

2. Core Functions and Technical Summary

The system implements the following core functions to meet the daily management needs of teachers:

- Student Information Management: Support adding new students with auto-generated unique student IDs, deleting students by ID with confirmation mechanism, and querying students by class or the entire grade.
- Score Management: Support updating student scores of Chinese, English and Math, with built-in score validity check to ensure the score is within 0-100, and display complete student score information in formatted table.
- Data Persistence: Automatically load existing data from local files when system starts, and save modified data to files to ensure data will not be lost after exit.
- Score Ranking: Support ranking by total score or single subject, for either the entire grade or a single class, to meet different statistical needs.
- Score Statistics: Automatically calculate the maximum, minimum, average score and pass rate of each subject, helping teachers quickly grasp the overall learning situation.

Technically, the project uses pure Python standard library, no third-party dependencies.

3. Application of OOP Concepts

This project fully implements all OOP concepts introduced in the course:

- Class and Object Abstraction: Core real-world entities are abstracted into classes. The Student class abstracts student entity with its attributes and methods, while the ScoreManager class abstracts the entire management system with all business logic. Each student is an instance of Student class, and the system itself is an instance of ScoreManager class, mapping real business to program structure.

- Encapsulation and Access Control: All attributes in Student class are defined as private attributes to avoid illegal direct modification. Standard getter and setter methods are provided to access and modify data, with validity check in setter methods (e.g. score range check) to ensure data integrity. This hides internal implementation and provides unified interface, conforming to OOP encapsulation idea.

- Modular Programming: The code is split into 3 independent modules by single responsibility principle: student.py for student entity definition, student_manager.py for business logic, main.py for user interaction. This clear structure allows independent development and test, convenient for later maintenance and expansion.

4. External Resources Declaration

This project does not use any third-party packages, external code or other non-original resources. All code is independently implemented by the developer, and only uses Python standard library os for file existence check. The project is not based on any previous course work, all content is original development for this course project.

5. Self-reflection and Future Improvements

Current Weaknesses:

- Interface Limitation: The system only provides command-line interface, not friendly enough for users unfamiliar with command line.
- Storage Limitation: Current storage uses plain text files, with poor scalability, no support for complex queries, and data conflict risk in multi-user scenario.
- Function Limitation: Only supports 3 fixed subjects, cannot dynamically add new subjects, and lacks advanced statistical functions.
- Exception Handling: Exception handling is not perfect, damaged data file may cause program crash directly.

Future Improvement Directions:

- Optimize the user interface
- Add input validation to prevent invalid user input.
- Expand functions to support dynamic subject addition and more statistical features.
- Add user permission management to ensure data security.
- Improve exception handling and data recovery mechanism to enhance system robustness.

6. User Guide

Running Steps: Place the 3 code files in the same directory, run python main.py in terminal .(we have provided data for 100 students, available for direct trial.)

Operation Instructions: After system starts, enter the corresponding number to select function:

- Show Students: View student information, support all students or specific class.
- Add Student: Add new student, system will auto-generate unique ID, input name and class.
- Delete Student: Delete specified student, input student ID and confirm.
- Update Score: Update student's scores, system will verify score validity automatically.
- Score Ranking: View score ranking, select scope and ranking basis.
- Score Statistics: View score statistics of the selected scope.

- Save and Exit: Save changes to local file and exit, or discard unsaved changes.

7. Repository Information
Github link: https://github.com/So-663/COMP2090SEF_Project_Group11
You can watch the video at github